HEMORA - Identification of Hematological Diseases Using Deep Learning and Microscopic Blood Smear Image Processing

25-26J-514

SLIIT
FACULTY OF COMPUTING

Introduction Video

HEMORA - Identification of Hematological Diseases Using Deep Learning and Microscopic Blood Smear Image Processing

Problem & Its Impact

- **Manual Microscopy:** Traditional peripheral blood smear examination by pathologists
- **Time Delay:** 3-7 days for results in government hospitals due to specimen transport and specialist availability
- **Human Error:** Inter-observer variability 20-35%, subjective assessment prone to misdiagnosis
- **Rural Access:** 80% of rural clinics lack hematology specialists

Research Objective

- **Main Objective**

  Develop a hybrid AI system capable of detecting:

  - Acute Lymphoblastic Leukemia
  - Iron Deficiency Anemia
  - Beta Thalassemia
  - Sickle Cell Disease

- **Key Approach**

  Combine:

  - Blood smear image analysis
  - CBC report parameter analysis

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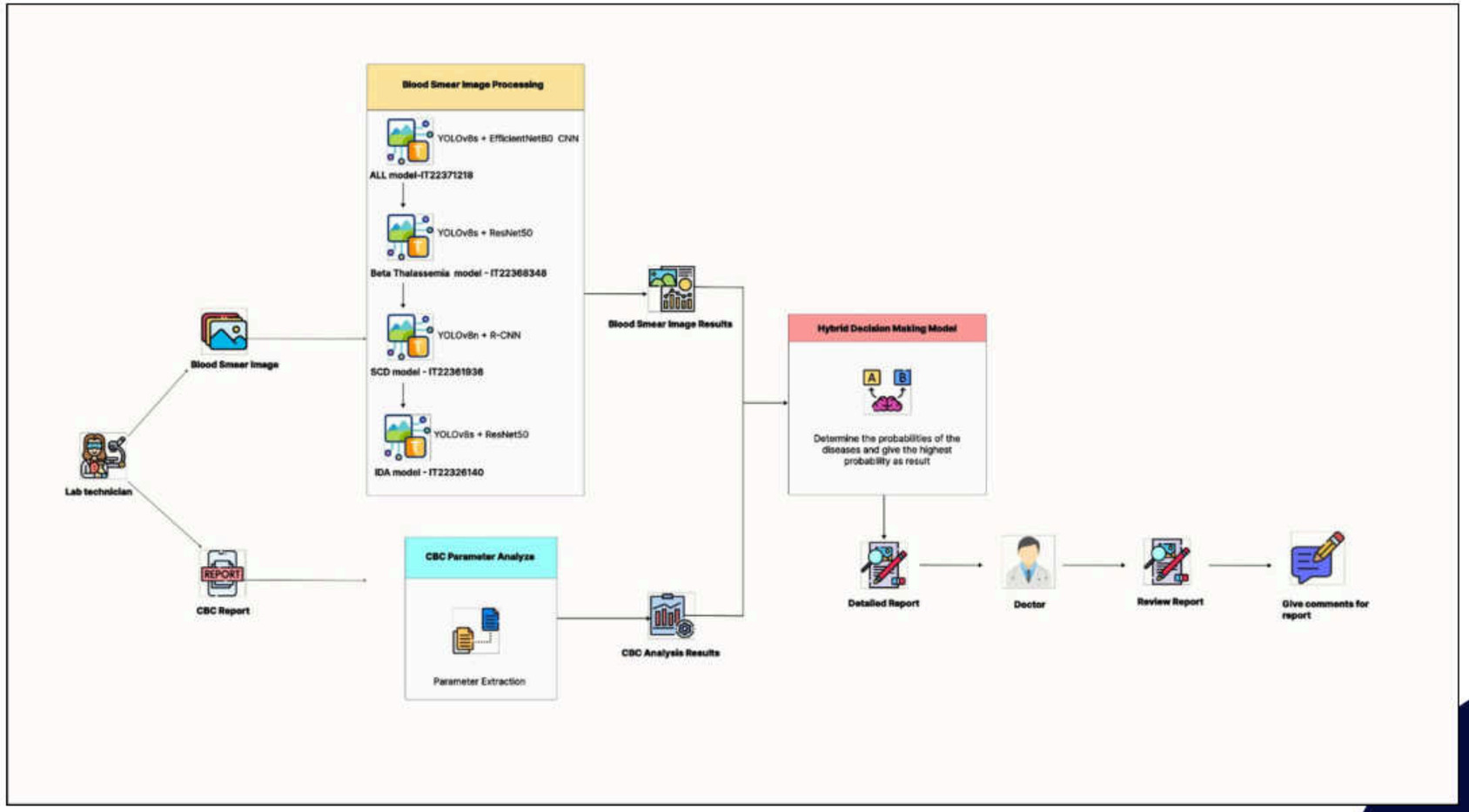
System Architecture & Integration



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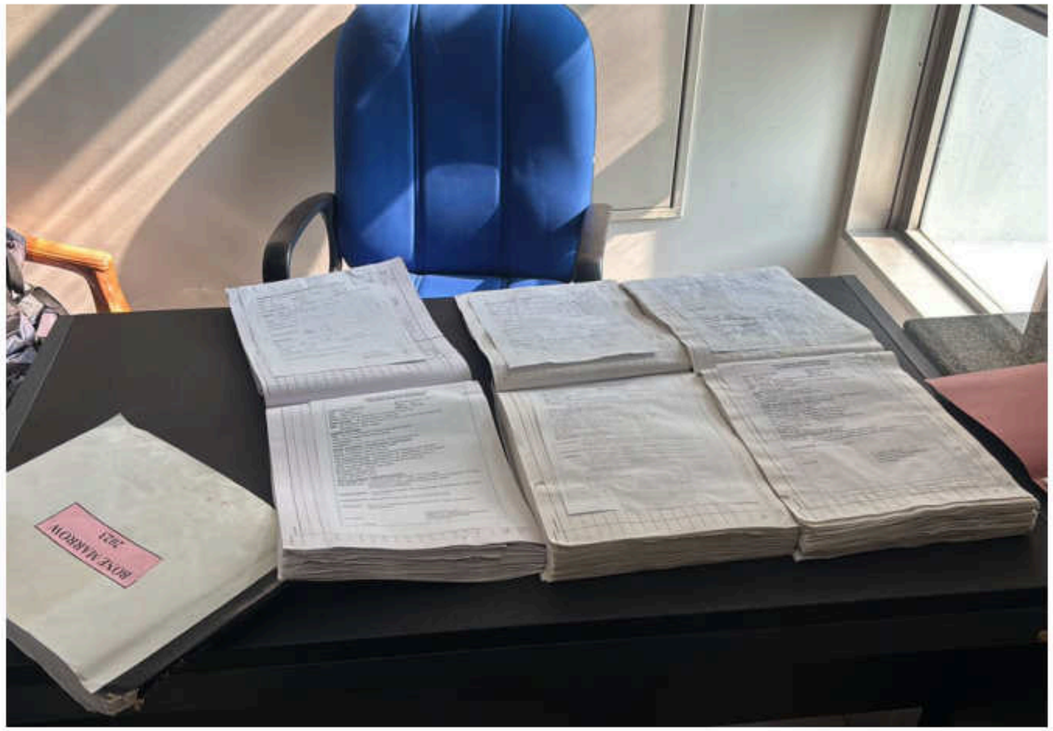
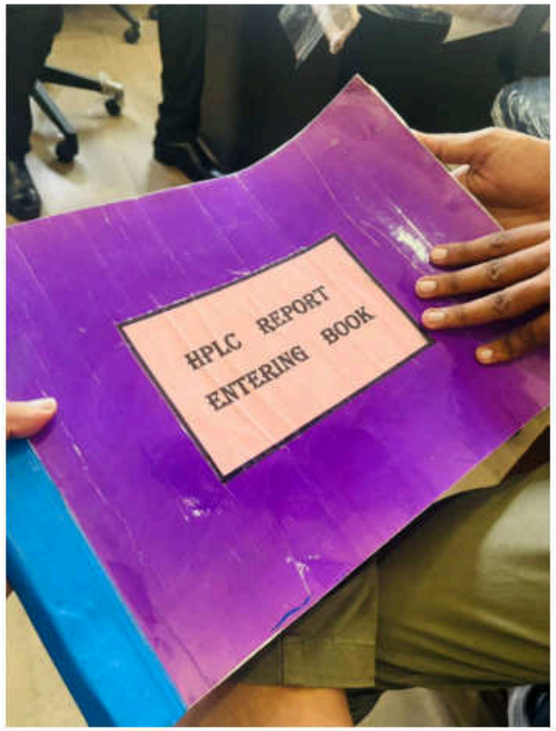
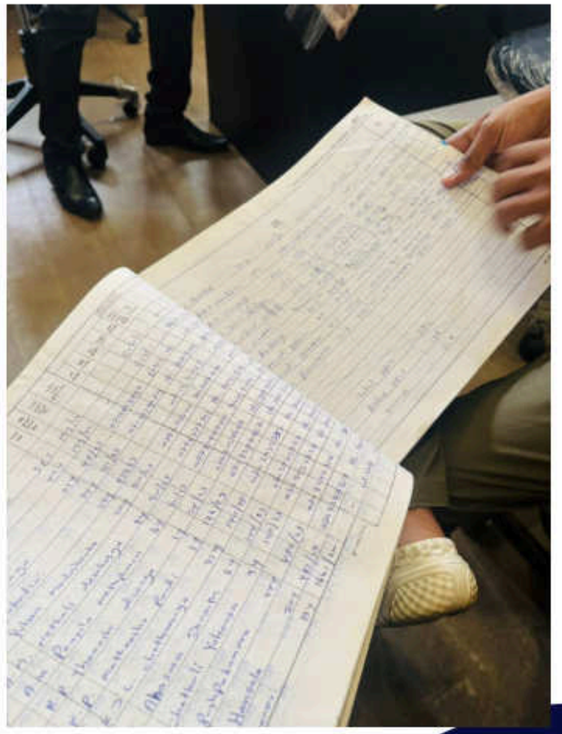
Data Collection

1) Referred to diagnosis reports and recorded the bone marrow IDs of patients diagnosed with the diseases







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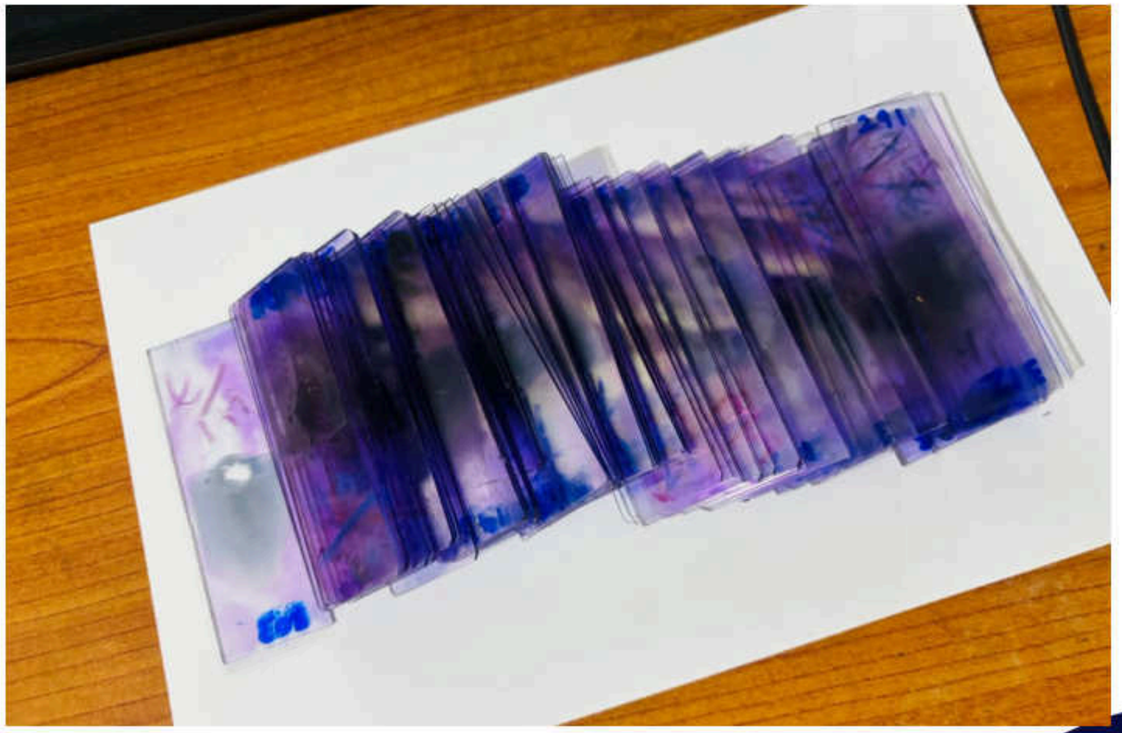
Data Collection

2) Collected peripheral blood smear slides using the recorded bone marrow IDs



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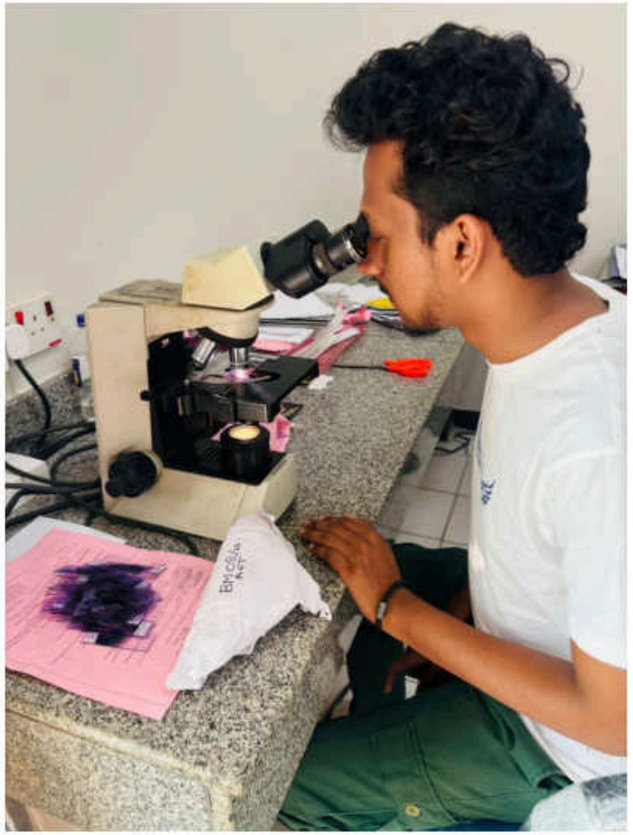
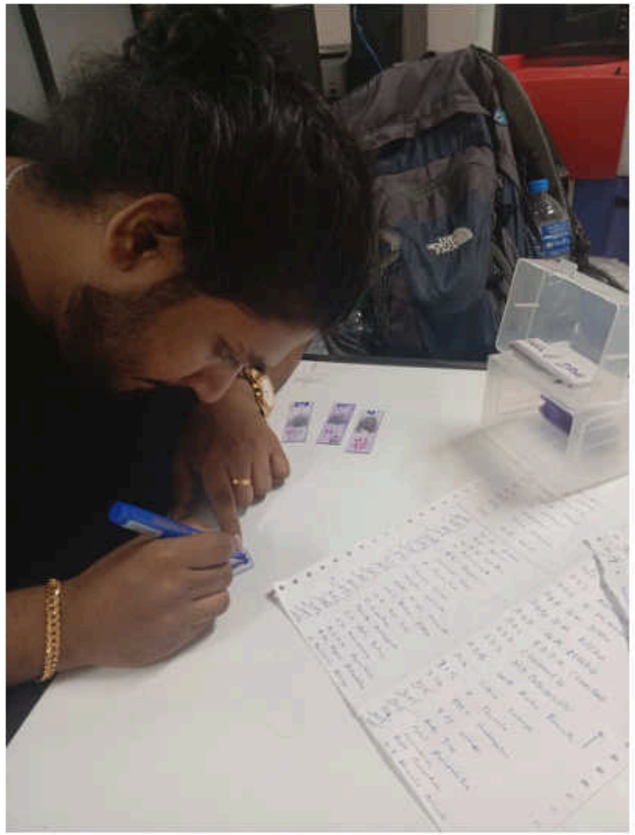
Data Collection

3) Selected suitable slides for the research dataset to capture images and identify features





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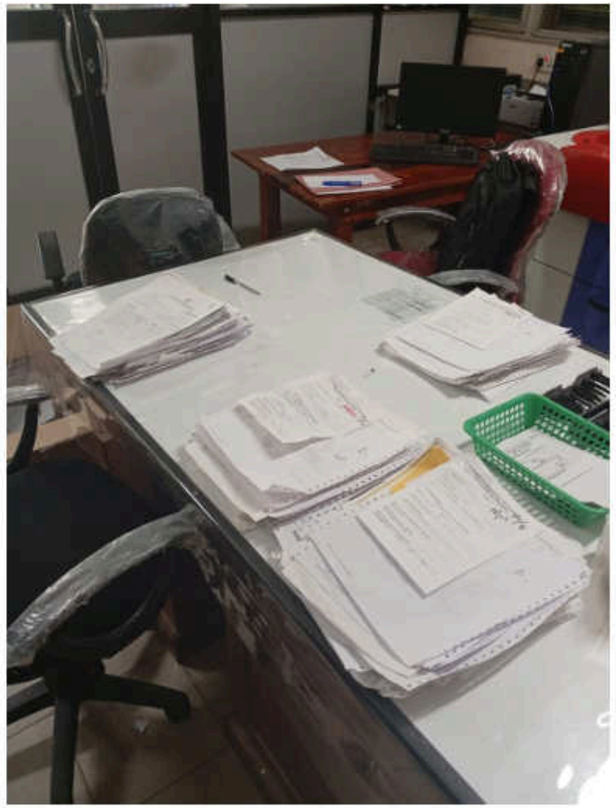
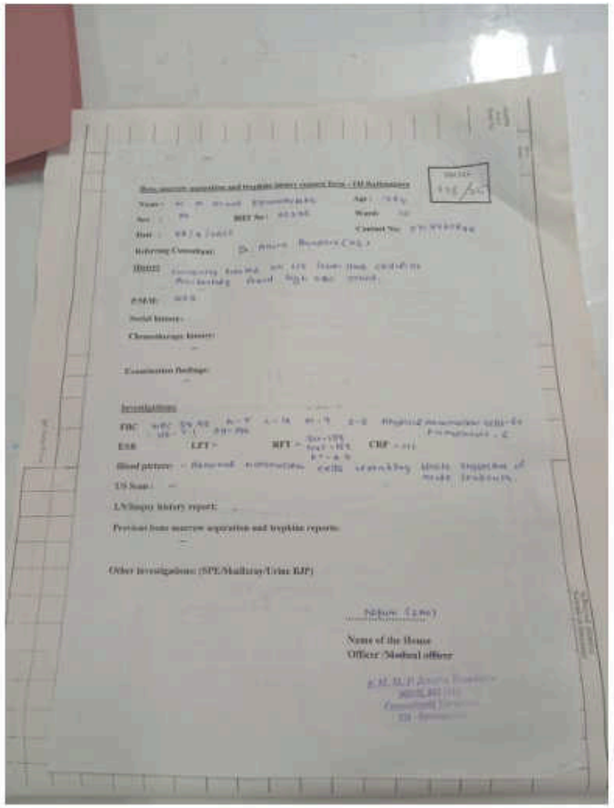
Data Collection

4) Collected CBC information related to the patients





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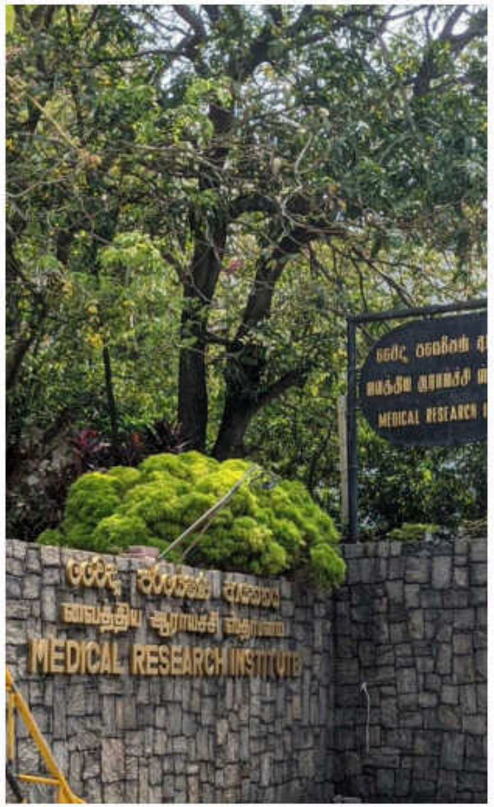
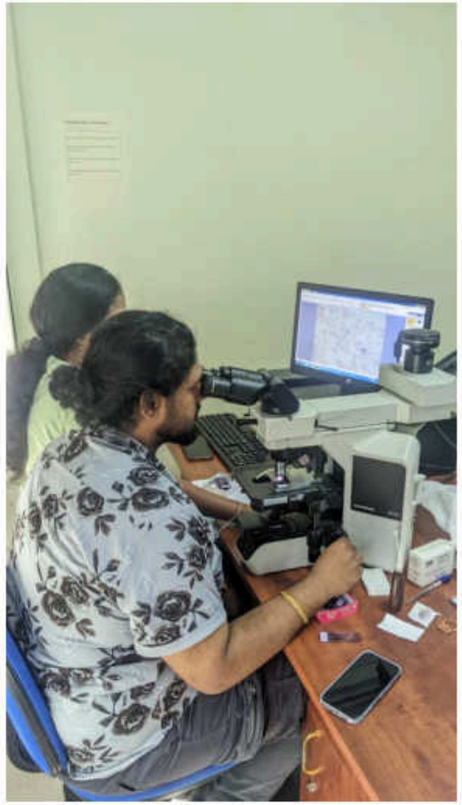
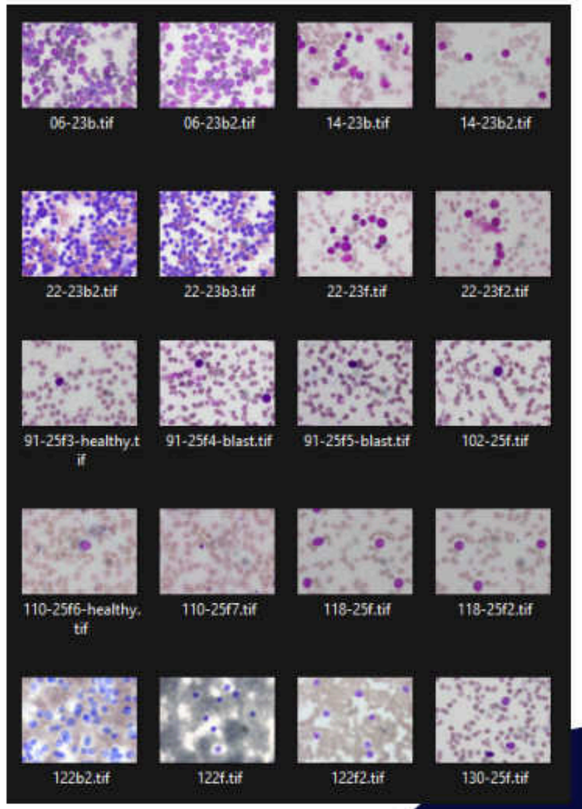
Data Collection

5) Captured blood smear images using a microscope connected to a computer to build a Sri Lankan-based dataset







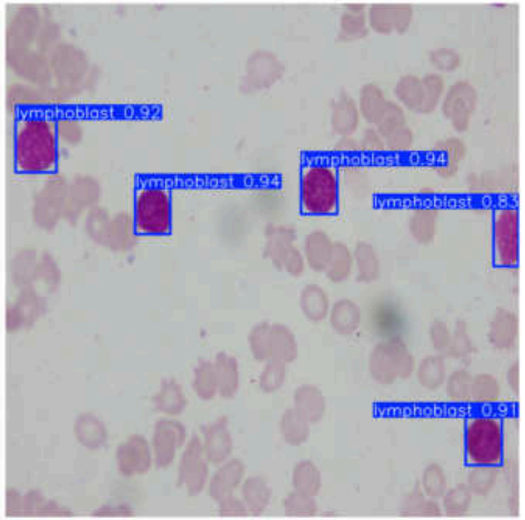
IT22371218 | Aluthge A.D.M.S

Detection and Identification of Acute Lymphoblastic Leukemia (ALL)

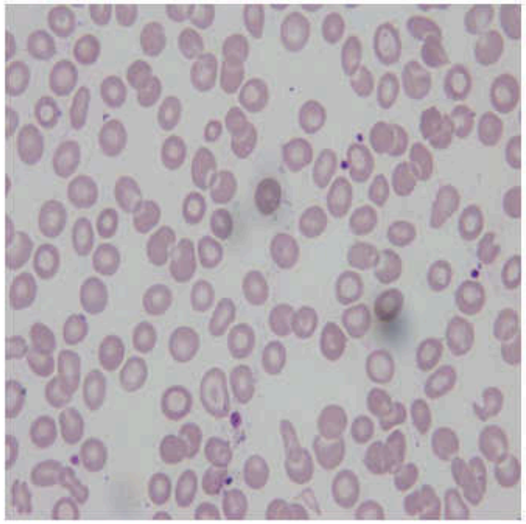
Specialization: Software Engineering

Detection and Identification of Acute Lymphoblastic Leukemia (ALL)

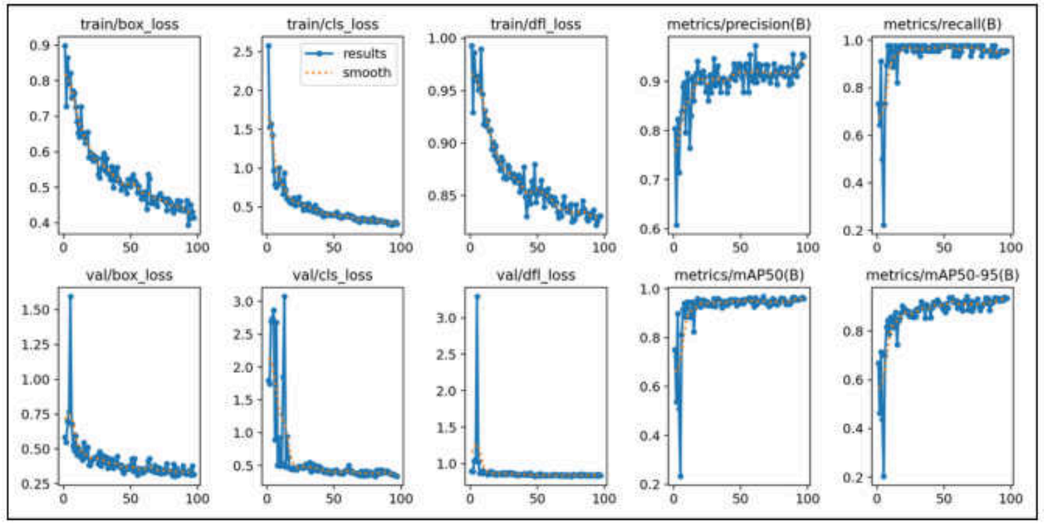
- After collecting the local dataset, retrained the YOLOv8s model to detect lymphoblasts



Lymphoblasts detected in an ALL-positive image

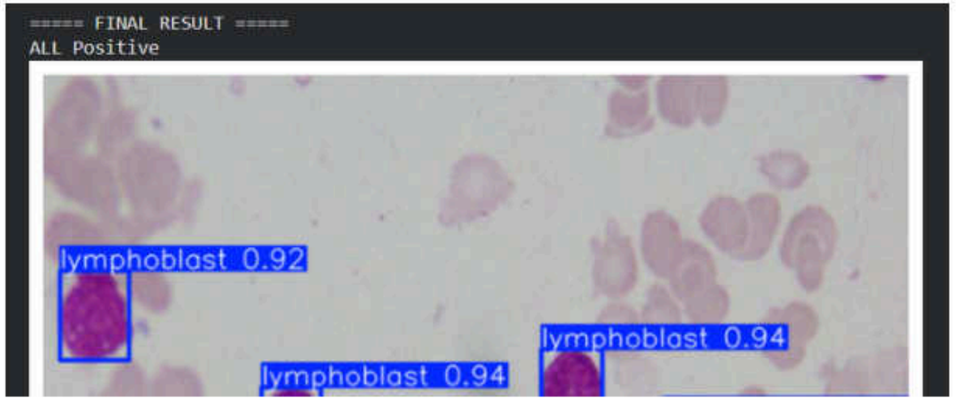


No detections in an ALL-negative image

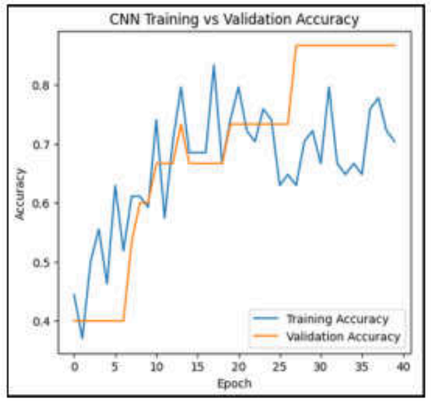


related graphs

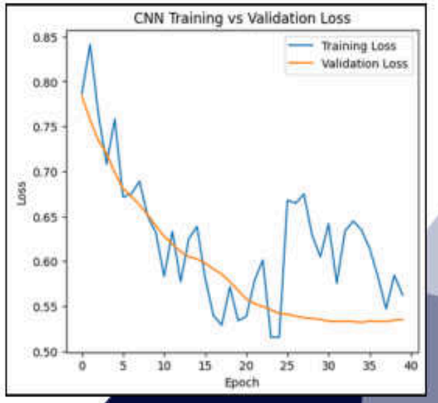
- Trained an EfficientNetB0 CNN model to classify images as ALL positive or ALL negative and combined with YOLO to make a hybrid model



Final Result



CNN accuracy graph



CNN loss graph

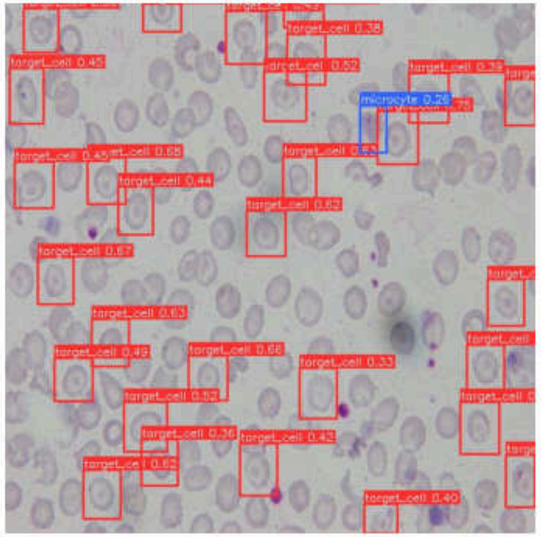
IT22368348 | Disanayake R.G.O.S

Detection and Identification of Beta-Thalassemia (Minor)

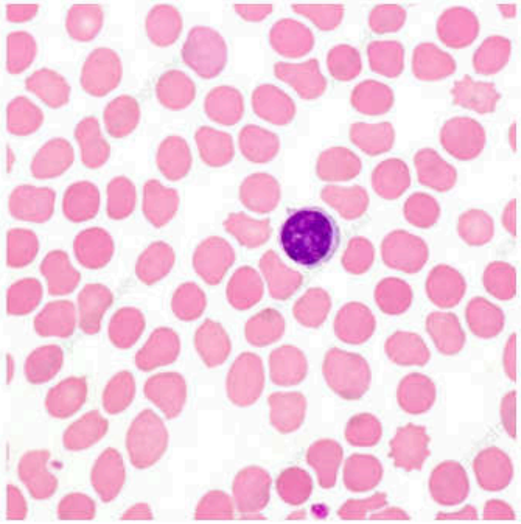
Specialization: Software Engineering

Detection and Identification of Beta-Thalassemia (Minor)

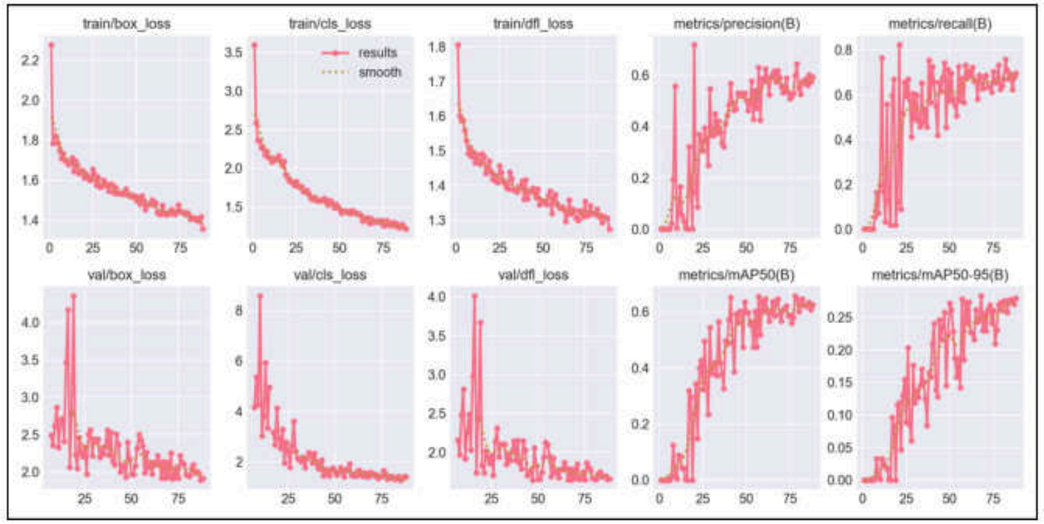
- After collecting the local dataset, retrained the YOLOv8s model to detect target cells, microcytes and hypochromia



Abnormal cells detected in an Thal-positive image

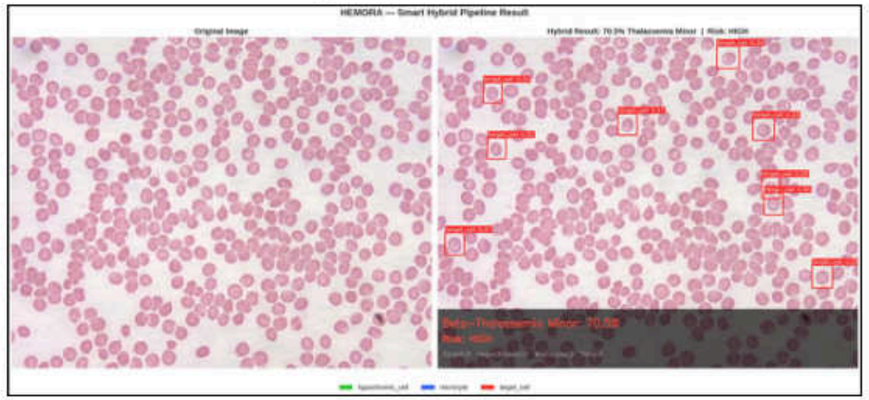


No detections in an Thal-negative image

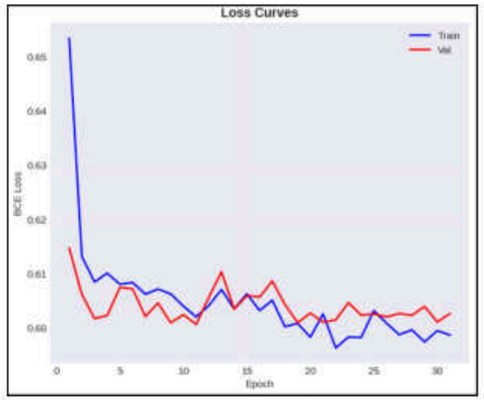


related graphs

- Trained a ResNet50 model to classify images as Beta Thalassemia positive or negative with a probability score, and combined it with YOLO to create a hybrid model



Final Result



ResNet50 Loss Curve

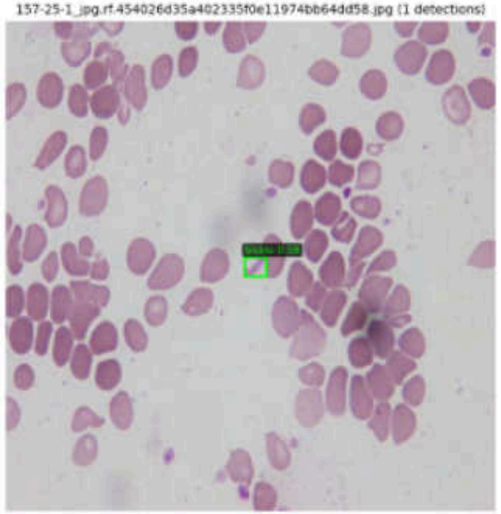
IT22361936 | Dassanayake E.G.N.L

Detection and Identification of Sickle Cell Disease (SCD)

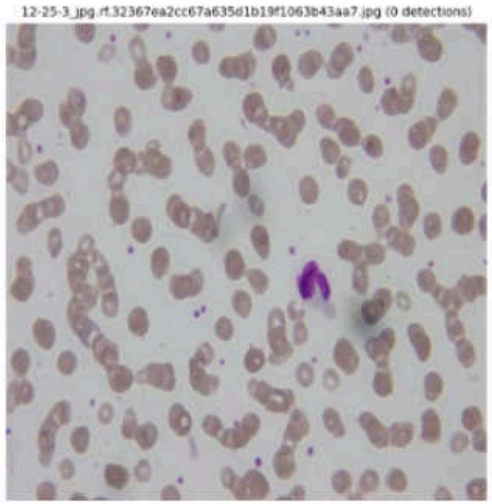
Specialization: Software Engineering

Detection and Identification of Sickle Cell Disease (SCD)

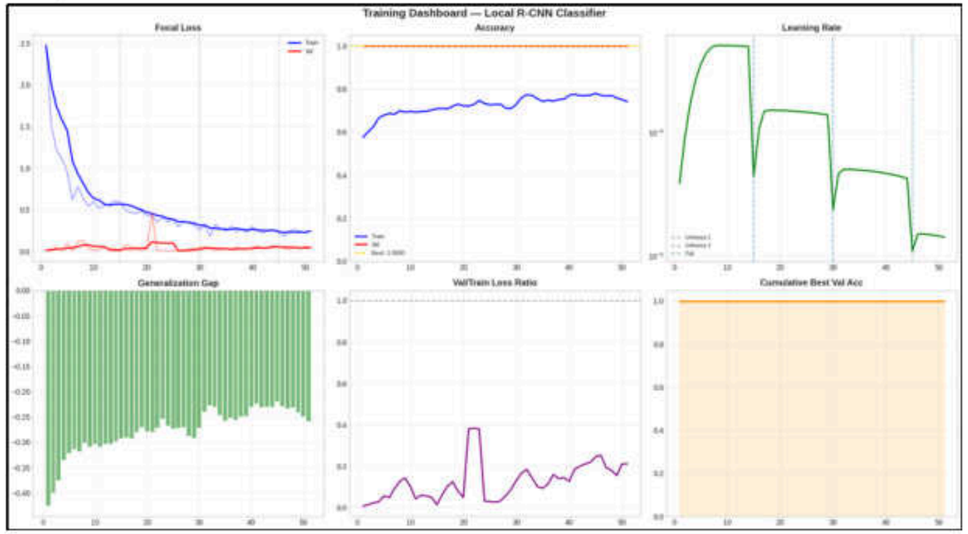
- After collecting the local dataset, retrained the YOLOv8n model to detect sickle shaped cells



Sickle cell detected in an SCD-positive image

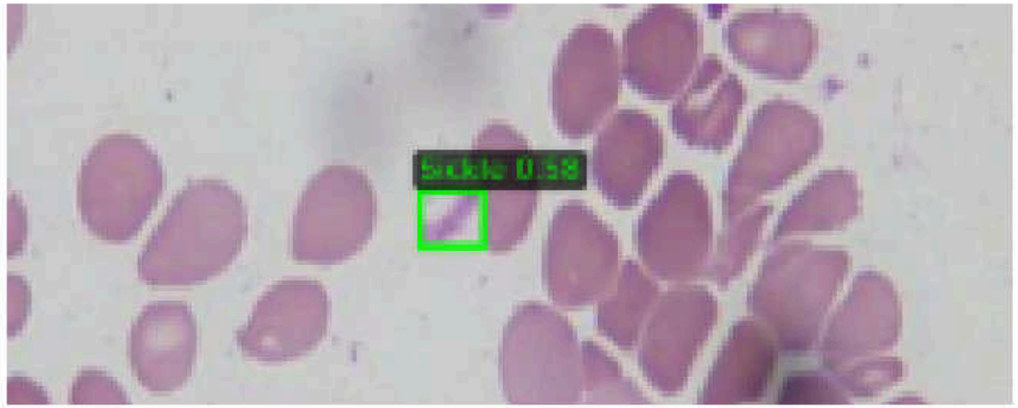


No detections in an SCD-negative image

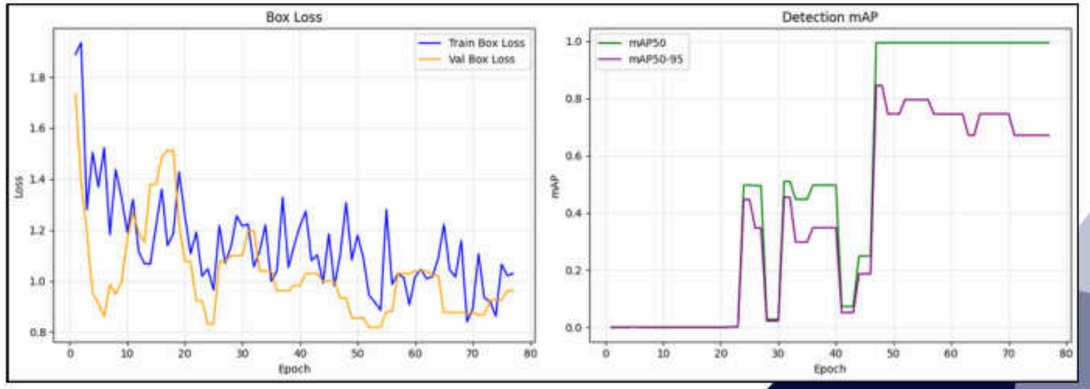


Related graphs

- Trained an R-CNN model to classify images as SCD positive or SCD negative and combined with YOLO model to make a hybrid model



Final Result



Train and validation loss graph

Detection Map

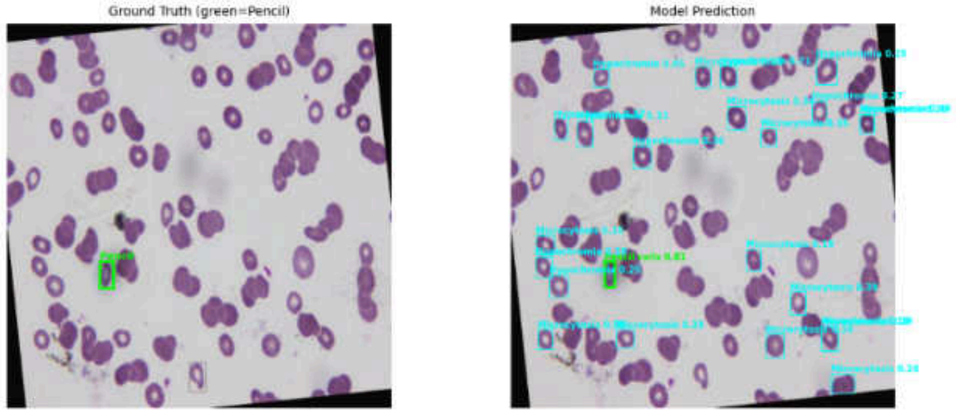
IT22326140 | Sachinthanee M.A.S

Detection and Identification of Iron Deficiency Anemia (IDA)

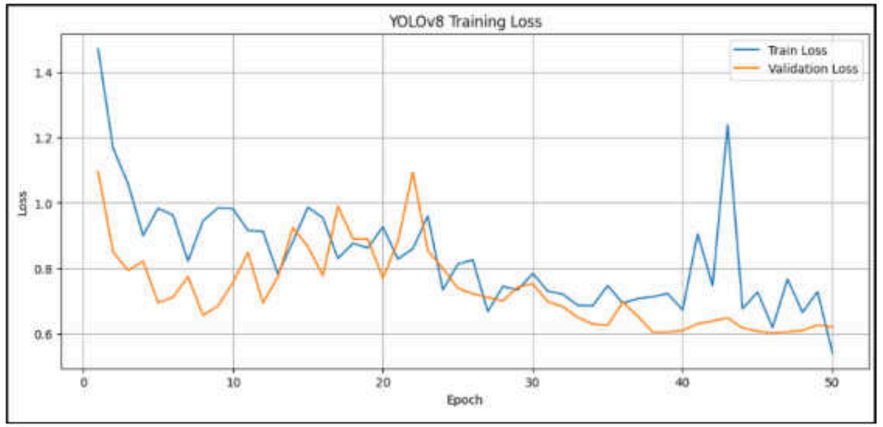
Specialization: Software Engineering

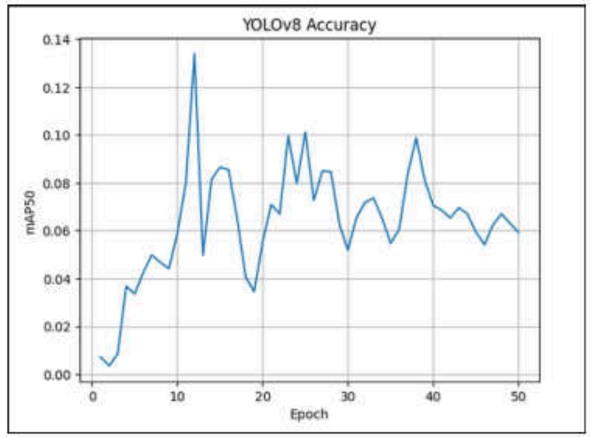
Detection and Identification of Iron Deficiency Anemia

- After collecting the local dataset, retrained the YOLOv8s model to detect pencil cells, microcytes and hypochromia



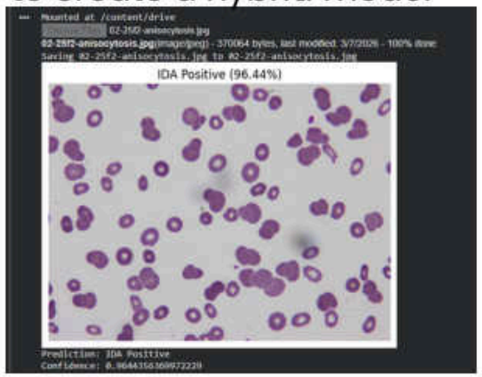
Abnormal cells detected in an IDA-positive image

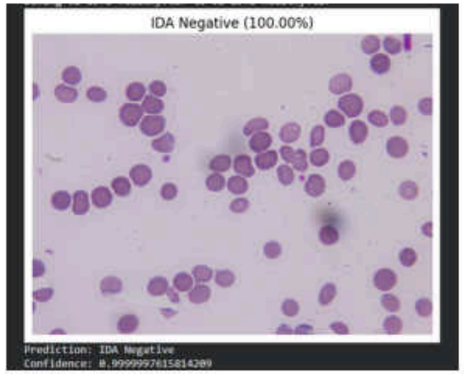




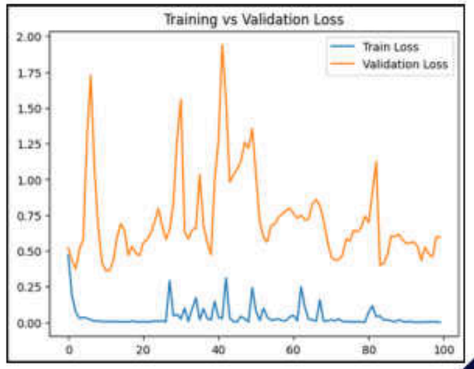
Related graphs

- Trained a ResNet50 model to classify images as Iron Deficiency Anemia positive or negative with a probability score, and combined it with YOLO to create a hybrid model





IDA image Results



ResNet50 Loss Curve

HEMORA – Identification of Hematological Diseases Using Deep Learning and Microscopic Blood Smear Image Processing

Contributions

IT22371218 | Aluthge A.D.M.S

- Collected blood smear slides and CBC information from Ratnapura General Hospital.
- Examined blood smear slides under a microscope and captured smear images at the Medical Research Institute.
- After collecting the local dataset, retrained the YOLOv8s model to detect lymphoblasts.
- Trained an EfficientNetB0 CNN model to classify images as ALL positive or ALL negative and combined with YOLO to make a hybrid model.
- Implemented the backend for the doctor comment section.
- Participated in deploying the system.

IT22368348 | Disanayake R.G.O.S

- Collected blood smear slides and CBC information from Ratnapura General Hospital.
- Examined blood smear slides under a microscope and captured smear images at the Medical Research Institute.
- After collecting the local dataset, retrained the YOLOv8s model to detect target cells, microcytes and hypochromia.
- Trained a ResNet50 model to classify images as Beta Thalassemia positive or negative with a probability score, and combined it with YOLO to create a hybrid model.
- Implemented CBC parameter extraction from uploaded CBC reports and integrated it with the system's decision layer.
- Participated in deploying the system.

HEMORA - Identification of Hematological Diseases Using Deep Learning and Microscopic Blood Smear Image Processing

Contributions

IT22361936 | Dassanayake E.G.N.L

- Collected blood smear slides and CBC information from Ratnapura General Hospital.
- Examined blood smear slides under a microscope and captured smear images at the Medical Research Institute.
- After collecting the local dataset, retrained the YOLOv8n model to detect sickle cells.
- Trained an R-CNN model to classify images as SCD positive or SCD negative and combined with YOLO to make a hybrid model.
- Integrated the AI models to the project.
- Participated in deploying the system.

IT22326140 | Sachinthanee M.A.S

- Collected blood smear slides and CBC information from Ratnapura General Hospital.
- Examined blood smear slides under a microscope and captured smear images at the Medical Research Institute.
- After collecting the local dataset, retrained the YOLOv8s model to detect pencil cells, microcytes , anisocytosis and hypochromia.
- Trained a ResNet50 model to classify images as IDA positive or negative with a probability score, and combined it with YOLO to create a hybrid model.
- Implemented the backend CRUD function for patient .
- Participated in deploying the system.

HEMORA - Identification of Hematological Diseases Using Deep Learning and Microscopic Blood Smear Image Processing

Solution Overview

Solution Summary

- Developed a hybrid AI-based diagnostic system that detects hematological disorders using peripheral blood smear images and CBC report analysis.

Non-Functional Focus

- **Performance:** Optimized models for fast prediction
- **Reliability:** Hybrid decision system improves diagnostic accuracy
- **Usability:** Simple interface for uploading smear images and CBC reports
- **Security:** Safe handling of uploaded medical data

Key Features Delivered

- Blood smear image analysis using deep learning models to detect disease-specific cell morphology.
- Automatic CBC report data extraction from uploaded report images.
- Hybrid prediction system combining image-based AI results with CBC parameters to generate the final diagnosis.

Current Completion Status

- >95% completed

HEMORA - Identification of Hematological Diseases Using Deep Learning and Microscopic Blood Smear Image Processing

Customer Persona

- **Customers:** Hospitals, diagnostic labs, NGOs, rural clinics
- **Users:** Hematologists, lab technicians, physicians

Market Size

- **Growing Digital Healthcare :** More hospitals adopt digital systems.
- **High Blood Disorder Prevalence :** Common among local patients.
- **Limited Specialists :** Few experts available.
- **High Patient Load** : Overcrowded public hospitals.

What Makes It Unique

- **Hybrid AI Model :** Uses multiple technologies to identify blood disorders.
- **Multi-Model Analysis :** Combines blood smear images and CBC reports for prediction.
- **Sri Lankan Dataset :** Models trained using local blood data and CBC parameters.

Cost Recovery

- Subscription for hospitals/labs (e.g., LKR 10,000 – 15,000 / month)
- Pay-per-use for clinics (e.g., LKR 200 – 300 / test)
- Possible government/NGO funding for scaling